# Krzysztof Mazurek

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## Summary

Enthusiastic about modeling, simulation, and control of dynamic mechanical systems, with interests in topics such as: GNC, mission analysis, multibody dynamics. I aim to deepen my expertise in numerical computing, software development and optimization.

## Education

**Politecnico di Milano** — Sep. 2025 - Present (expected - 2027)
*Master's degree in Space Engineering*

**Warsaw University of Technology** — Oct. 2021 - Mar. 2025
*Robotics and Automatic Control - Final grade : 4.4 / 5*

## Professional experience

**Structural Analysis Engineer** — May 2025 - Aug. 2025
*Sener Aerospace & Defence* — *Warsaw, Poland*

- Performed static and modal FEA of structures (ESA COMET container, ARIEL satellite components) using Siemens Femap.
- Automated FEA workflows using VBA and Python: temperature mesh mapping from Esatan to Femap, result postprocessing, midsurface preprocessing.

**Software Engineer (Internship)** — Jul. 2024 - Aug. 2024
*Sellita Watch Company* — *Le-Chaux de Fonds, Switzerland*

- Developed software module for Beckhoff PLC in TwinCat3- movement control of ultra-precise servo press for assembling watch components based on the user-specified criteria.
- Developped HMI for the machine control module in C#.Net framework.

## Student activities

**Multibody and simulation software engineer** — Nov. 2025 -
*Polispace - Rover Team* — *Milano, Italy*

- Developing the model of the rover in Simscape and its integration with Unreal Engine.

**Electrical team coordinator, Mechanical team member** — Jan. 2022 - Jun. 2025
*Student Association for Vehicle Aerodynamics - MotoStundent Electric 2025* — *Warsaw, Poland*

- Responsible for early-stage development of the electric drivetrain- design and validation.
- Simulated and optimized shape of a motorcycle rear pro-link suspension in Msc Adams.
- Designed a steel-tubed swingarm, chain tensioner assembly, rear brake adapter using Siemens NX.

**3D printing Science Club vice-president** — Oct. 2022 - Oct. 2024
*Warsaw University of Technology* — *Warsaw, Poland*

- Designed and manufactured models, customized and assembled machines (CNC, resin/ FDM printers).

## Other projects

**Spacecraft Attitude Dynamics course project** — Sep. 2025 - Dec. 2025

- Attitude control of a microsat equipped with CMG, IMU, Sun & horizon sensor using LQR and nonlinear control, S/C dynamics and environment disturbances modeling - Simulink.

**Orbital mechanics course project** — Sep. 2025 - Dec. 2025

- Interplanetary fly-by mission design; Orbit perturbations modeling and analysis- Matlab.

**Thesis - Feedforward control of underactuated multibody systems in a co-simulation framework** — Oct. 2024 - Mar. 2025

- Developed a simulation environment for planar control of underactuated mechanisms using servo-constraints. It combined Python software communicating with MSC Adams via Simulink.

**Satellite solar array opening simulation** — Mar. 2024 - Jun. 2024

- Collaborative project with Politecnico di Milano for multibody dynamics course; using Msc Adams and Matlab.

## Technical Skills & Foreign language

**Programming**: Matlab & Simulink, Python, C++
**Engineering Software**: Msc Adams, Siemens Femap, Ansys Mechanical & Spaceclaim, Siemens NX, Fusion 360
**Tools**: Git, GitHub, Linux, LaTeX, MS Office, TwinCat3
**Foreign language:** English C1+, German B1, Polish native